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Development and Evaluation Of A Low-Cost Reference-Aware UV Fluorescence Inspection Architecture For PCB Conformal Coating

Scientific Literature Review and Experimental Methodology

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Data confidentiality. The company retains proprietary ownership of PCB images and production process parameters.

Abstract

Conformal coating is the process of applying a coating to a PCB to protect it from natural elements. To ensure that this process has been carried out properly, machine vision and external UV fluorescent light at a certain wavelength can be used. However, several factors can affect the inspection outcome, including optical filtering, material composition, shadows, ultraviolet detection using CNNs or other models, illumination, and angles. Consequently, using a CNN model with a camera cannot be considered novel. Instead, this research aims to produce a synthesis of evidence and standards within a cost-effective architecture for a fixed 70 mm printed circuit board assembly. This will involve corrections, geometry input, probabilistic coating output scores, and morphological defects. Experiments will use classical, learned, and hybrid techniques, together with hardware on the actual board, as well as the creation of a private dataset containing defects and defect-free examples.

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1. Research problem, objectives and boundaries

1.1. Industry status

The main use of this technology is to protect PCB core components, such as ICs, capacitors, and resistors, from natural elements such as moisture and other potential contaminants that can pose a reliability risk. This process can be inspected with vision systems; however, conformal coating is transparent, allowing ordinary light to pass through and making inspection complicated. To perform this test, ultraviolet fluorescent light can reveal the coating solution. It is important to note that fluorescence itself cannot estimate coating thickness or other parameters such as curing state, long-term protection, or adhesion. These matters are addressed in IPC-HDBK-830A for processing, coverage, and inspection, and in IPC-A-610J for assembled-board acceptability criteria, which include thickness and whole-board coverage specifications [1, 2, 3].

In this specific scenario, a board of about 70×70 mm, populated with several smaller PCBs, is positioned in a fixed position. The camera should target an area of 80×80 mm, allowing a 5 mm margin. Ideally, the camera is mounted in a fixed location, although its position may shift in the future. Commercial systems currently add oblique cameras for specific PCB areas [4, 5].

1.2. Research objectives

The main goal of this work is to assess the possibility of developing a low-cost UV fluorescence station with fixed geometry and image analysis to inspect conformal coating on a 70×70 mm PCB panel. The expected KPIs are camera response and excitation; designing a board coverage map from CAD; comparing segmentation under identical conditions; correctly identifying defect types; and auditing CPU latency and cost.

To make a scientific claim, a comparator must be included. Several previous studies have conducted UV bubble inspections using classical methods and CNNs [6, 7]. Other studies have claimed that various features can be statistically classified using YOLOv8 tiling and UV intensity [8]. Detector networks have also been trained for dewetting and capillary defects [9]. Therefore, neither hybrid processing nor CNN use for conformal coating is novel. In this case, novelty arises from isolating properties and setting actual constraints for defect handling, illumination robustness, and uncertainty in human review.

2. Literature review

2.1. Mechanisms of protection and material variability

Conformal coating refers to a thin film that conforms to the geometry of the electronic assembly. IPC-HDBK-830A describes acrylic, epoxy, silicone, polyurethane, and parylene coating systems and related applications, curing, compatibility, coverage, and thickness [1]. These chemical systems vary in physical properties and processing requirements, even within a family. Formulations, fluorescent tracer amounts, and fluorescence spectra can also vary. It is crucial to include the production coating data sheet and qualification results as experimental inputs. A coating that does not produce adequate fluorescence upon 365 nm exposure disqualifies the chosen imaging method and should prompt a design review rather than attempts to improve unreliable images.

A functional inspection taxonomy should distinguish between physical phenomena and image appearance. Missing coating refers to a lack of the required coating. Overspray or keep-out refers to the presence of coating in an unpermitted location. A bubble indicates trapped gas in or under the coating; a blister could refer to separation of the coating from the substrate. Dewetting results in local coating recession. Capillary pooling involves accumulation of coating around leads and geometrical features. Specific criteria for cracks, wrinkles, scratches, and contamination may need to be developed together with an alternative lighting approach. These classes are candidates for annotation rather than necessarily detectable classes. IPC-A-610J defines coverage and thickness classes for conformal coating and, in its 2024 version, made changes related to coating voids/bubbles [2, 10].

2.2. Excitation, emission and optical contrast

In a simple first-order model, the signal can be written

$$I(\mathbf{x}) \approx g\,t\,q(\mathbf{x})\,E_{365}(\mathbf{x})\,\eta(\mathbf{x}) + B(\mathbf{x}) + \varepsilon(\mathbf{x}), \tag{1}$$

where g is camera/system gain, t the exposure, q the combined camera/lens/filter response, E_{365} local excitation irradiance, η an effective emission response of tracer and coating geometry, B residual background/reflections and ε noise. Equation 1 is an engineering abstraction. In particular, η may depend nonlinearly on film thickness and concentration.

2.3. Field of view, physical sampling and resolution

With an 80 mm horizontal FOV and a 4000-pixel image width, image sampling equals

$$s_x = \frac{80\,\mathrm{mm}}{4000} = 0.020\,\mathrm{mm/pixel} = 20\,\mu\mathrm{m/pixel}. \tag{2}$$

A 0.20 mm geometric feature covers about 10 pixels horizontally in the ideal projected image.

Image width	Sampling (μm/px)	0.10 mm feature (px)	0.20 mm feature (px)
1600 px	50.0	2.0	4.0
2448 px	32.7	3.1	6.1
3264 px	24.5	4.1	8.2
4000 px	20.0	5.0	10.0

2.4. Lighting geometry and experimental controls

Component heights cause bright specular regions, self-shadowing and occlusion. A top-only flood source may illuminate PCB areas, yet miss coating on vertical component faces. Side UV sources can recover some information; commercial designs use both top and oblique views [4, 5]. The direct Choi et al. study is pertinent: it uses brightness/hue features, Mahalanobis distance and YOLOv8 tiling to evaluate UV intensity for bubble detection [8].

2.5. Thickness versus fluorescence and independent validation

Shao et al. measured the interfaces of conformal coatings non-destructively using ultra-high-resolution optical coherence tomography [11].

2.6. Safety constraints for optical development

365 nm UVA can present photobiological hazards to eyes and skin. IEC 62471 provides classification and measurement concepts for optical-radiation hazards from lamps including LEDs [12]; ISO 12100 sets out machine risk assessment and risk-reduction principles [13]. Neither a nominal LED optical wattage nor a wavelength alone determines safe exposure.

3. Direct prior art and industrial inspection systems

3.1. Direct academic research on coated PCBs

The two 2021 Korean Journal of Broadcast Engineering papers by Lee et al. directly address conformal-coated PCB bubble inspection [6, 7]. In the earlier approach, UV images were processed to find dark-centre candidates, after which CNN-based verification rejected non-bubble lookalikes. The authors' abstract reports precision 0.805, recall 0.763 and F1 0.767. This discrepancy is precisely why the original evidence matters. In the later paper, VGG-style CNN transfer learning is reported to improve bubble detection to F1 0.9044.

Zouhri and colleagues presented a Faster R-CNN assessment of coating bubbles at ICM 2021 [14].

In 2024, Choi et al. addressed UV irradiance selection for small conformal-coating bubbles, using image-level colour/brightness descriptors, feature-space distances, YOLOv8 and image tiling [8]. Wang et al. reported a YOLOv8-Nano variant with attention and feature fusion for PCB coating bubble detection, with a 12% relative performance improvement for smaller bubbles in the institutional abstract [15].

Patel et al. presented a customised Faster R-CNN with ResNet-152 and enhanced proposal mechanisms for capillary and dewetting patterns on conformal-coated PCBAs [9]. The abstract reports recall of 98.53% and 95.49% for two defect categories, respectively.

3.2. Relevant work

General PCB AOI surveys discuss sensor/illumination choices, geometric comparison, segmentation, classification and manufacturing integration [16, 17, 18]. The 1988 Hara et al. system exploited ultraviolet-excited *PCB base-material fluorescence* to distinguish copper patterns [19].

Outside electronics, Han et al. used autoencoders, self-labelling, and convolutional models to identify coating boundaries on fuel-injection nozzles [20]. Their fully accessible paper reports over 95% interface localization within ± 6 pixels and 7.18 images per second.

3.3. Commercial conformal-coating AOI systems

CAD-driven forbidden/required regions, UV illumination and multiple camera views cannot individually be claimed as new. Commercial product brochures rarely provide a publicly reproducible dataset with independent board counts and false-acceptance operating points.

Table 1: Published manufacturer descriptions. Numeric specifications are vendor claims, not independent head-to-head benchmarks.

Supplier / product	Relevant published capabilities	Engineering implication
Nordson FX-940UV [21]	5 MP colour image, UV illumination, taught known-good coating/noncoating areas, optional oblique views; manufacturer describes inspection in a few seconds.	Golden reference plus rules are established practice; 5 MP can be commercially relevant for some configurations.
Göpel Multi Line CCI [4]	Top and side UV, CAD/coating-plan regions, claimed 9/17 μm sampling, optional multiple oblique cameras and separate thickness measurement.	Region semantics and geometric coverage plans are established; complex 3D areas demand more than one view.
MIRTEC GENESYS-CC [5]	50 MP camera, claimed 4 μm optics and 30 μm microbubble sensitivity; AI-based bubble assessment, additional UV and side camera geometry.	Defect size and inspection risk determine resolution; published claims need independent validation.
Viscom CCI [22]	UV-triggered fluorescence for coating presence/evenness; distinguishes optical inspection from spot thickness measurement.	Treat fluorescence intensity as a process signal, not calibrated thickness without metrology.

3.4. Synthesis

The reviewed commercial pages demonstrate use of reference plans, but do not independently establish how much they improve each specific defect class.

4. Algorithms, public datasets and methodological trade-offs

4.1. Classical vision is a necessary baseline

For a single known PCB design under controlled UV illumination, classical vision offers a strong, interpretable starting point. Correct dark frames, gain and spatial irradiation; align to fiducials or stable geometric features. Otsu's histogram thresholding is a principled unsupervised baseline [23]. Geometric alignment can use fiducials and a planar projective transformation with robust RANSAC fitting [24]; camera intrinsics and lens distortion can be estimated with planar calibration [25]. Image registration reviews distinguish feature-based from intensity-based matching and the risks from resampling/occlusion [26].

4.2. CNN and transformer segmentation

U-Net remains a reasonable low-data segmentation comparator because its skip connections support preservation of spatial detail [27]. DeepLabv3+ uses atrous processing and a boundary-refining decoder, offering a more complex reference model [28]. Lightweight backbones such as MobileNetV3 may reduce CPU cost [29].

Swin Transformer aggregates hierarchical contextual representations [30]; SegFormer pairs multi-scale features with a small MLP decoder [31]; Mask2Former supports multiple mask segmentation tasks [32]; MobileViT investigates low-resource global representation [33]. The real evaluation criterion here is not a published score on ADE20K. A practical progression is (1) a classical reference-aware baseline; (2) small U-Net; (3) SegFormer-B0 or MobileNetV3-based segmentation if justified.

For very small bubbles, downsampling a full 4000-pixel image to 512 or 640 pixels wide changes effective sampling from 20 to 125–156 μm/px. Consequently, a nominal 0.2 mm defect that spans ten original pixels may span only about 1.3–1.6 pixels after aggressive resize. The Choi et al. paper is direct evidence that tiling has been investigated for tiny coating bubbles [8].

4.3. Anomaly detection and scarce defects

PatchCore represents acceptable image patches with a feature-memory bank [34]; DRAEM trains with synthetically generated anomalies [35]; EfficientAD explores a low-latency student–teacher arrangement and global context [36].

The uncertainty-aware PCB method U^2D^2PCB illustrates reconstruction and discriminative anomaly logic on conventional PCB data [37]. Likewise, CM-UNetv2 targets small ordinary PCB features and boundaries, offering a hypothesis about feature preservation rather than evidence of coating behavior [38].

4.4. Dataset inventory and actual relevance

Dataset	Scale / access	What it contains	Use in present study
DeepPCB [39]	1,500 template/test pairs; source released	Conventional PCB pattern defects, six categories	Registration/detection pipeline only
DsPCBSD+ [40, 41]	10,259 images, 20,276 boxes; Figshare CC BY	Copper residue/deficiency, scratches and foreign objects; local crops	Annotation tooling, small-object benchmark
FPIC [42]	Research collection described in preprint	Semantic optical PCB assurance images	General semantic PCB task exploration
MVTec AD [43, 44]	5,354 images; noncommercial license	15 object/texture categories, anomalies	Anomaly-code sanity check only
MVTec LOCO AD [45]	3,644 images, noncommercial	Structural and logical anomalies	Test reference-rule ideas conceptually
VisA [46]	10,821 images, CC BY	Industrial anomalies including PCB-like categories	Transfer learning or anomaly comparison, not coating ground truth

DsPCBSD+ provides explicitly downloadable images and object annotations under an official Figshare record [41]. This makes it useful for validating image loaders, annotation conversion and neural training code, but those image labels refer to bare PCB surface/copper defects. DeepPCB has reference/test image pairs and defect boxes, which could help test registration software before any UV data are captured [39]. MVTec AD and LOCO AD carry non-commercial dataset licences; they must not silently migrate into a factory training or redistribution workflow [44, 45]. VisA lists a more permissive CC BY dataset licence but is still not a substitute for collecting UV coating images [46].

4.5. Dataset acquisition is a central experimental contribution

A purpose-built UV coating dataset is therefore a *necessary deliverable* and possibly an independent scholarly contribution. It should include nominal 70 mm boards across physical board IDs, coating lots, component configurations and controlled acquisition conditions; carefully separated genuine defects and induced test coupons.

4.6. Prior-work synthesis matrix

Table 2: Directness of the principal academic comparisons.

Study	Primary relevant contribution	Evidential constraint
Lee et al. 2021a/b [6, 7]	Coating bubble UV candidate extraction plus CNN / transfer learning	Defect type primarily bubbles; no demonstrated production transfer to our line
Choi et al. 2024 [8]	UV level suitability, multitype bubble location and tiled YOLOv8	Does not settle other coating defects or CPU runtime
Wang et al. 2025 [15]	Enhanced YOLOv8-Nano for small PCB coating bubbles	Relative improvement reported; detailed sample counts unverified
Patel et al. 2026 [9]	Faster R-CNN for dewetting/capillary coating defects	Class recalls without general production false-acceptance operating point
Han et al. 2025 [20]	Self-labeling coating-interface localization	Fuel-injection nozzles, not UV PCB surfaces
Shao et al. 2019 [11]	Independent non-destructive coating thickness metrology	Hardware and observable different from fluorescence coverage

5. Proposed reference-aware inspection architecture

5.1. Design rationale and data flow

A constrained board-specific coating plan defines *where* coating is required or forbidden; the optical model measures *whether* sufficient fluorescence evidence is observed; and an optional learned branch estimates ambiguous coating structure or defect patterns. Figure 1 presents the proposed data flow.

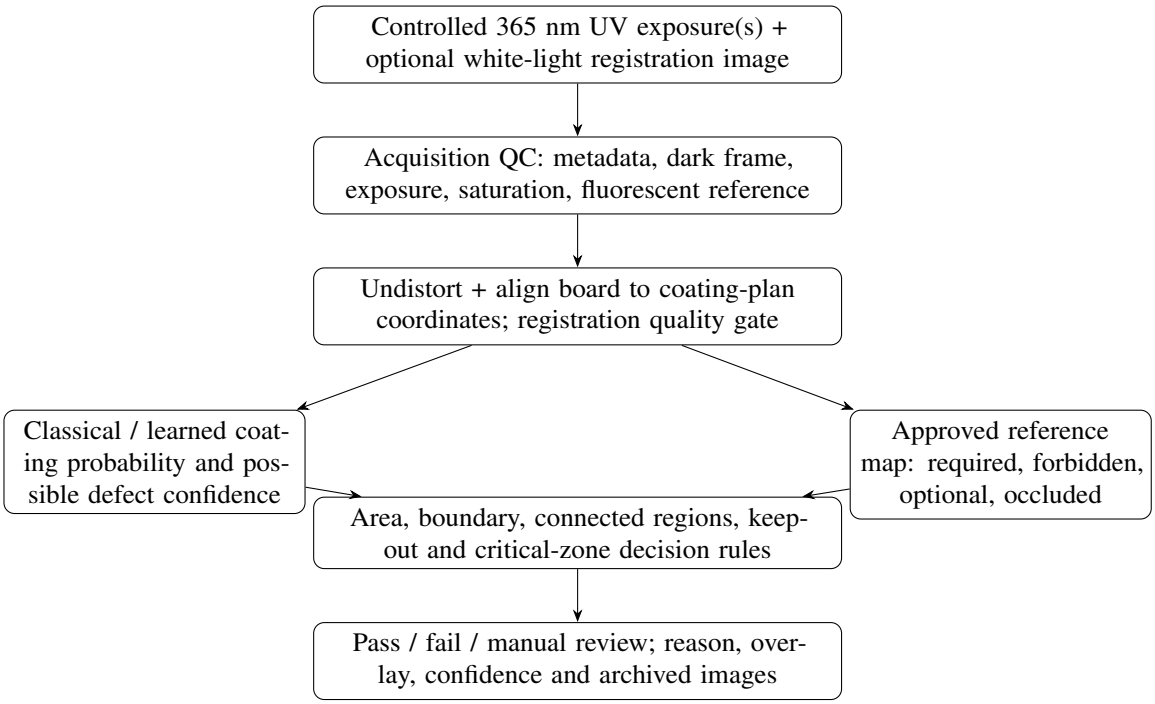


Fig. 1. Hypothesized inspectable and auditable architecture.

5.2. Radiometric correction and quality gate

For an illuminated raw image I with dark frame D and stable flat fluorescent reference F imaged under the same conditions:

$$I_c(\mathbf{x}) = \frac{I(\mathbf{x}) - D(\mathbf{x})}{\max(F(\mathbf{x}) - D(\mathbf{x}), \varepsilon)} \overline{(F - D)}. \tag{3}$$

Here ε prevents division by near-zero values and $\overline{(F-D)}$ controls overall scaling. Correction is accepted only where the reference is stable, correctly registered and not clipped. It *does not* unshadow occluded surfaces or compensate for all changes in the fluorescent tracer's chemistry. Quality gates flag saturation, insufficient brightness, excessive dark-frame residual, missing board and out-of-focus image. A failed acquisition gate must produce *review/reacquire*, not an automatic PASS.

5.3. Registration and reference coordinates

Let $\mathbf{p}$ denote an image pixel and $\mathbf{q}$ its location in the coating-plan coordinate frame. A planar projective transform $\mathbf{q} \sim H\mathbf{p}$ can be estimated from fiducials or stable features using RANSAC [24, 25].

Approved mask regions form a disjoint partition of the visible board: R = must-coat, K = must-not-coat, A = optional/allowed, and U = not reliably inspectable/occluded. Every pixel belongs to exactly one region after resolving drawing overlaps.

5.4. Segmentation output and interpretable indicators

Define $P(\mathbf{x}) \in [0,1]$ as estimated probability or standardized evidence of coating, and $M_\tau(\mathbf{x}) = \mathbb{1}[P(\mathbf{x}) \geq \tau]$ as a binary mask chosen on validation data.

$$C_R = \frac{|M_\tau \cap R|}{|R|}, \qquad C_K = \frac{|M_\tau \cap K|}{|K|}. \tag{4}$$

These are *image-derived ratios*, not coating-thickness measurements. Overall coverage may conceal a critical local hole; therefore identify connected missing regions $H_i \subset R \setminus M_\tau$, measure each area $A_i = n_i s_x s_y$, maximum inscribed radius, contour length and proximity to critical features.

$$D = \begin{cases} \text{REVIEW}, & Q = 0 \text{ or uncertainty} > u_{\max}, \\ \text{FAIL}, & \exists \text{ validated actionable region violating } R \text{ or } K \text{ rules}, \\ \text{PASS}, & \text{otherwise within defined visible inspection scope}, \end{cases} \tag{5}$$

where Q summarizes acquisition, registration and model validity. A PASS must be explicitly scoped to inspected visible surfaces and validated defect categories.

5.5. Model selection and ablation logic

The segmentation branch should be replaceable: a calibrated HSV/intensity threshold, lightweight U-Net, MobileNetV3 decoder, SegFormer-B0 or a learned anomaly score. The reference-comparison branch must use identical region rules and optical inputs where possible across experiments. Ablations should remove individual modules one at a time: (A) flat-field correction; (B) geometric registration; (C) the learned mask; (D) the reference regions; (E) multi-angle illumination; and (F) uncertainty handling.

6. Reproducible experimental methodology

6.1. Specimen and lot sampling

Each physical PCB must have a unique immutable `specimen_id`; each production/process batch has `lot_id`. Every image receives an `image_id` linking it to coating material, preparation, light arrangement, camera parameters and board orientation.

An initial *planning target* could be approximately 100–200 physical boards across at least three visually distinct collection sessions or lots.

6.2. Defect taxonomy and acceptance thresholds

Table 3: Candidate annotation and decision ontology. Inspection feasibility and acceptance limits are to be established, not assumed.

Class	Label representation	Independent confirmation / concern
Missing coating	Pixel polygon or region instance	UV evidence plus expert/alternate inspection; distinguish intentional mask-out
Excess coating / overspray	Pixel region crossing forbidden zone	Coating plan revision and expert confirmation
Bubble / void	Instance region, size and location	Microscopy or expert adjudication; fluorescent dark spots are not all bubbles
Dewetting	Boundary polygon, area and context	Process engineer confirmation; often resembles absence or capillary effect
Capillary pooling	Region plus component proximity	Fluorescence intensity alone may not quantify thickness
Crack / scratch	Instance or line segment	Alternate-angle white/UV imaging; distinguish surface artefacts
Occluded / not observable	Exclusion mask	Must be handled as unknown, not labelled sound coating

A defect is *actionable* only if a documented customer, internal drawing or authorized acceptance specification makes it rejectable under the applicable product class. Record minimum size, permitted locations, any exceptions for component leads and acceptable bubble conditions. The goal is traceable compliance with a validated inspection plan, not the maximum number of bright/dark pixels detected. The full IPC-A-610J and coating plan must be consulted by a suitably qualified person before production rules are finalized [2, 10].

6.3. Classical baseline A

Register the board geometry using fiducial-driven affine/projective fitting and fail closed on poor registration [24].

6.4. Learned baseline B

Record fixed random seeds, model input resolution, weight initialization, class balance, augmentation, loss definition, optimizer, learning rate and early stopping.

6.5. Production timing and system verification

The complete cycle includes trigger-to-exposure, optional multiple exposures, camera transfer, acquisition validation, geometric registration, any inference tiling, defect aggregation, logging/database output, PLC result handshake and safety/fault reporting. Measure median, 95th and 99th percentile times across sustained runs, along with cold start, model reload, CPU load, RAM, thermal throttling and recovery after camera disconnection. A nominal inference-only speed on a GPU-equipped laptop cannot be copied to a used Intel office PC. The workstation's actual takt requirement must be supplied by production engineering and the system benchmarked against it [47, 48].

7. Statistical analysis, reliability and production feasibility

7.1. Primary and secondary outcomes

The proposed *primary* outcome is board-level sensitivity for *actionable* defects at a predeclared maximum false-reject rate on independent boards. For a binary decision, define defect-positive ground truth, true positives (TP), false negatives (FN), false positives (FP) and true negatives (TN) at the *board level*. Then

$$\text{Sensitivity} = \frac{TP}{TP+FN}, \quad \text{Specificity} = \frac{TN}{TN+FP}, \quad \text{Precision} = \frac{TP}{TP+FP}, \quad \text{FPR} = \frac{FP}{TN+FP}. \tag{6}$$

7.2. Failure-mode and risk analysis

Table 4: Potential failure modes and the proposed monitoring response.

Failure	Consequence	Control / observable
LED power or spectral drift	Weak fluorescence appears as missing coating	Stable reference target, drift alarm, locked settings
Wrong/uncalibrated emission filter	Reflected UV dominates signal	Spectral check, reference subtraction, qualification records
Board misregistration	False keep-out failures or masked true flaws	Fiducial residual threshold, manual review state
Specular component/occlusion	Missing sidewall or dark inaccessible areas	Exclusion/occlusion map, optional side views
Coating batch variation	Shift in nominal intensity and defect texture	Lot-stratified validation, recalibration, OOD alarm
Model/rule mismatch	Silent false pass after PCB revision	Versioned CAD mask, software/configuration compatibility checks
Dataset leakage	Inflated laboratory accuracy	Physical-board and lot keyed splits, audit script
Camera/PC failure	Missing inspection or stale result reuse	Trigger sequence IDs, watchdog, fail-safe PLC handshake

7.3. Standards and deployment validation

IPC material qualification (IPC-CC-830C), assembled-board acceptance (IPC-A-610J) and process guidance (IPC-HDBK-830A) play different roles [3, 2, 1]. Optical and mechanical safety engineering should incorporate IEC 62471 and ISO 12100, with actual measurements and formal assessment [12, 13]. Attribute-based lot sampling concepts can support ongoing process monitoring (ISO 2859-1:2026), but must not be conflated with proving sensitivity of the automated 100%-inspection device [49].

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